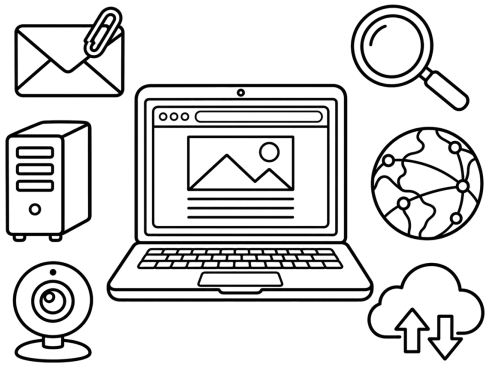
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 11



UNIT 3
The Internet and the Electronic Mail

Student Workbook

"The King of ICT"

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BSc (Sp) Computer Science - University of Peradeniya
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MIT (Reading) - University of Sabaragamuwa

STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain what the Internet is and who looks after it.
- [ ] Name the parts of a URL, an IP address and a domain name.
- [ ] Explain how the DNS, the protocols and the client server model work.
- [ ] Name the services of the Internet and use a browser and a search engine.
- [ ] Send an e-mail with Cc, Bcc, a subject and an attachment.
- [ ] Explain instant messages, video conferencing and social networks.
- [ ] Describe cloud computing: IaaS, PaaS and SaaS.

HOW TO USE THIS BOOK

Read each chapter with your teacher, do the practical tasks and then write the exercises. The **WRITING SPACES** are for the notes your teacher gives you in class - write them neatly: they are your revision notes for the O/L examination.



THE WORDS YOU WILL LEARN

Internet	URL	HTTP	IP address
ISP	Domain name	Top Level Domain	DNS
Protocol	TCP/IP	Client	Server
Downloading	Uploading	World Wide Web	Web browser
Hyperlink	FTP	Streaming	Search engine
E-mail address	Cc / Bcc	Spam	Cloud computing
Home page	Remote access	Video conference	Social network

"The Internet puts the whole world on your desk. Use it to learn, not only to play."

- THE KING OF ICT

"Stay connected to your goals, even when the signal is weak."

- THE KING OF ICT

CHAPTER 1

The Internet and Its Addresses

? THE PAGE THAT WOULD NOT OPEN

"The exam results web site does not open!"

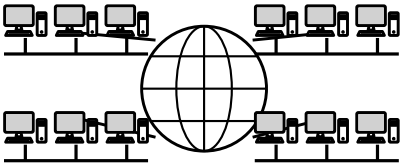
"You typed doenets.k - you missed the l of lk!"

Every part of a web address has a job. Learn what the **Internet** is, and how a **URL**, an **IP address** and a **domain name** find every web site and every computer.



3.1 What is the Internet?

The **Internet** is a collection of **computer networks around the world**. It is the fastest way to share information with the world (the **information super highway**), and it has made the whole world a **global village**.



The Internet has **no single owner**. A non-profit organisation, **the Internet Society**, is in charge of the ethics and principles of using the Internet and of its **protocols**. A computer that connects to the Internet also becomes a computer of the Internet.

3.2 The Uniform Resource Locator (URL)

Information is a very important **resource**. The protocol **HTTP** (Hypertext Transfer Protocol) carries it between the computers of the Internet. The information of the **World Wide Web (WWW)** is kept in **web sites**, and the **Uniform Resource Locator (URL)** uniquely identifies each resource in a web site.

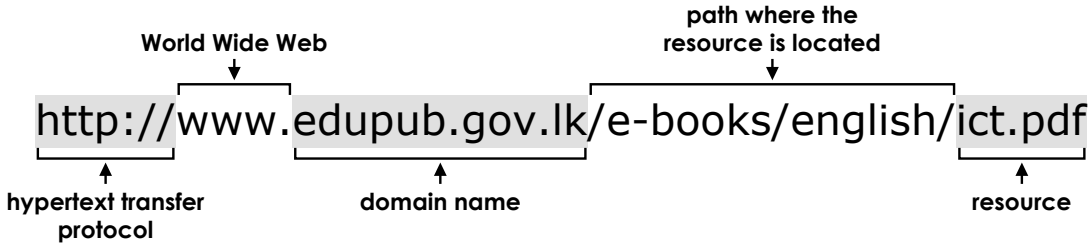


Figure 3.1 - Uniform Resource Locator

Note: A URL must have all the parts **from the HTTP protocol to the domain name**. Some URLs of the field of education:

http://www.moe.gov.lk Ministry of Education	**http://www.nie.lk** National Institute of Education
http://www.doenets.lk Department of Examinations	**http://www.edupub.gov.lk** Educational Publications Department

WRITING SPACE 1 - Parts of a URL write what your teacher tells you

3.3 IP addresses

Internet Protocol (IP) addresses identify every computer in the Internet individually. An IP address has **four values from 0 to 255**, separated by a decimal point (a dot): this is **dotted decimal notation**, e.g. 172.64.85.42 and 193.213.78.154.

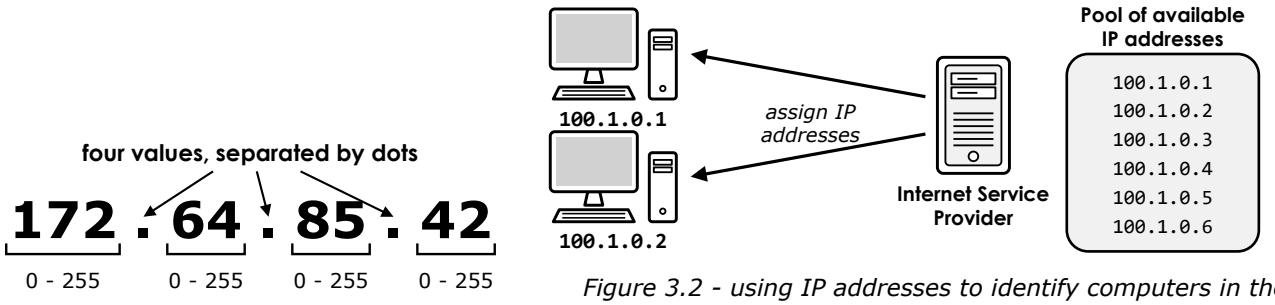


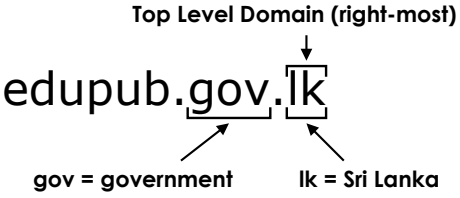
Figure 3.2 - using IP addresses to identify computers in the Internet

IP addresses are given by an **Internet Service Provider (ISP)**: a company that provides Internet and electronic mail facilities and charges fees for them.

WRITING SPACE 2 - IP addresses write what your teacher tells you

3.4 Domain names

In the domain name **edupub.gov.lk** (Figure 3.1), **gov** stands for the government and **lk** for Sri Lanka. The lk domain, which comes after gov, is the **higher** domain. A domain name gives every web site an identity like this.



"Every web site has an address. Every goal needs one too - write yours down."

- THE KING OF ICT

The domains of the fields

Domain	Meaning
com	Commercial
org	Non profitable organisations
gov	Government (public)
edu	Education
net	Web

The domains of the countries

Domain	Country
lk	Sri Lanka
in	India
au	Australia
jp	Japan
uk	United Kingdom
us	United States

The domain at the **right-most** end of a domain name is the **Top Level Domain**: com in google.com, lk in edupub.gov.lk and nie.lk, org in unicode.org.

DO IT - PRACTICAL 1 (TEXTBOOK PAGES 65 - 67)

Visit the education web sites of section 3.2 and find the IP address of your computer. **Ask your teacher first and do not change any setting.**

1	Open a web browser. Type **http://www.moe.gov.lk** and press **Enter**.
2	Find the protocol, the www and the **domain name** in the address bar.
3	Open **www.nie.lk**, **www.doenets.lk** and **www.edupub.gov.lk** in the same way.
4	Open an e-book on edupub.gov.lk. Find the **path** and the **resource** in its URL.
5	**Start** → **Settings** → **Network & Internet** → **Properties**: find the **IPv4 address**.
6	Check its four values (0 - 255). Compare it with a friend's IP address.

My IP address _____ My friend's IP address _____

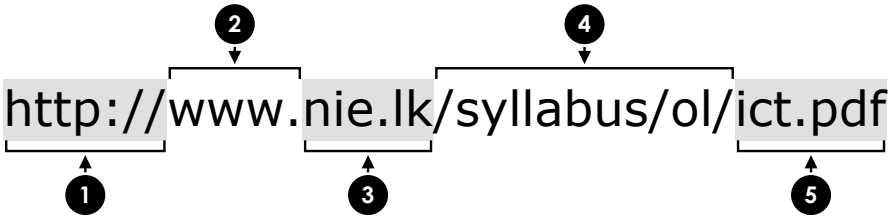
A Label the URL

5 marks

Write the name of each numbered part. Use the words in the box.

WORD BANK

domain name	hypertext transfer protocol	path where the resource is located
resource	World Wide Web	



1. ..... 2. .....

3. ..... 4. .....

5. .....

B Correct IP address?

6 marks

Tick one box for each IP address.

IP address	Correct	Not correct
172.64.85.42	☐	☐
193.213.78.256	☐	☐
209.191.122.70	☐	☐
100.1.0	☐	☐
74.125.20.103	☐	☐
192.168.2.10.1	☐	☐

C Find the domain names

8 marks

Write the domain name and the Top Level Domain of each URL.

URL	Domain name	Top Level Domain
http://www.moe.gov.lk		
http://www.yahoo.com		
http://www.doenets.lk		
http://www.pinterest.com		

D True or false

6 marks

Write T or F in the box.

1. ☐ One big company is the owner of the Internet.
2. ☐ The Internet is called the information super highway.
3. ☐ An IP address has four values separated by dots.
4. ☐ In the domain name edupub.gov.lk, gov is the Top Level Domain.
5. ☐ An Internet Service Provider charges fees for its facilities.
6. ☐ A URL can leave out its domain name.

CHAPTER 2

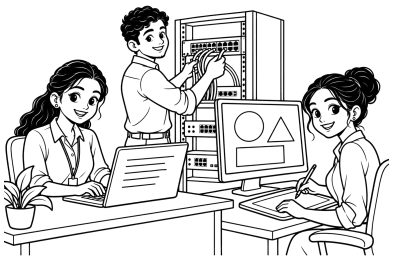
How the Internet Works

? NAMES FOR PEOPLE, NUMBERS FOR COMPUTERS

"How does my browser find www.yahoo.com?"

"It gets the IP address from a DNS server."

Learn how the **DNS**, the **protocols** and the **client server model** make the Internet work.



3.5 The Domain Name System (DNS)

People find it hard to remember lots of similar numbers; **names** are easier and cause fewer errors. So when a URL is typed in a web browser, the request must first be **translated into an IP address**. The **Domain Name System (DNS)** does this translation.

Example: a request for **www.yahoo.com** is first sent to the **DNS server**, which translates it into the IP address **209.191.122.70**. The real connection is made with this IP address, behind the screen, so quickly that the user does not notice.

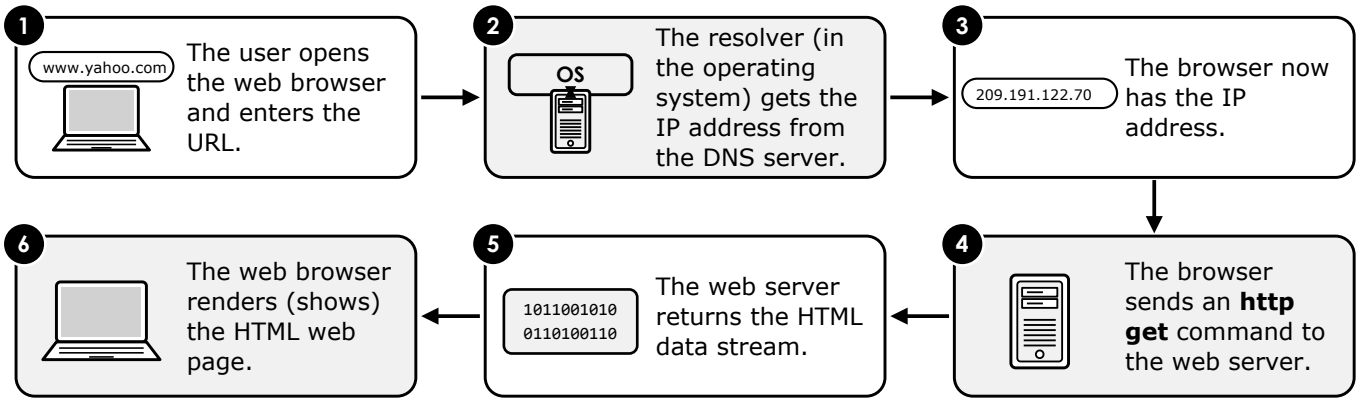
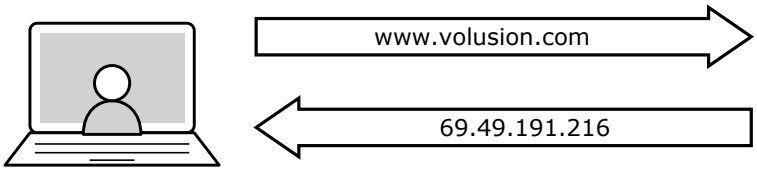


Figure 3.3 - the process of how Internet operations are executed

A **domain name** uniquely identifies a **web site** and is easy to remember; an **IP address** uniquely identifies a **computer**. The **Domain Name Server** converts the domain name into the IP address.



DOMAIN NAME SERVER

DOMAIN NAME	IP ADDRESS
www.volusion.com	**69.49.191.216**
www.google.com	74.125.20.103
www.facebook.com	173.252.110.27

Figure 3.10 - domain names are converted to IP addresses by the Domain Name Server

WRITING SPACE 3 - The Domain Name System write what your teacher tells you

WRITING SPACE 4 - How a web page opens write what your teacher tells you

3.6 Protocols

The Internet has no owner; it is controlled by **protocols**. A protocol is like a **system of law** in computer networks.

Protocol	Full name	Usage
HTTP	Hypertext Transfer Protocol	HTML document exchange
TCP/IP	Transmission Control Protocol / Internet Protocol	Controlling the exchange of IP addresses
FTP	File Transfer Protocol	File exchange
SMTP	Simple Mail Transfer Protocol	Electronic mail exchange
ICMP	Internet Control Message Protocol	Informing and controlling messages when there are errors

WRITING SPACE 5 - Protocols write what your teacher tells you

3.7 The client server model

The Internet is a **Wide Area Network (WAN)** based on the **client server model**: every computer of the Internet is a **server** or a **client**. A server distributes the resources that the client computers need.

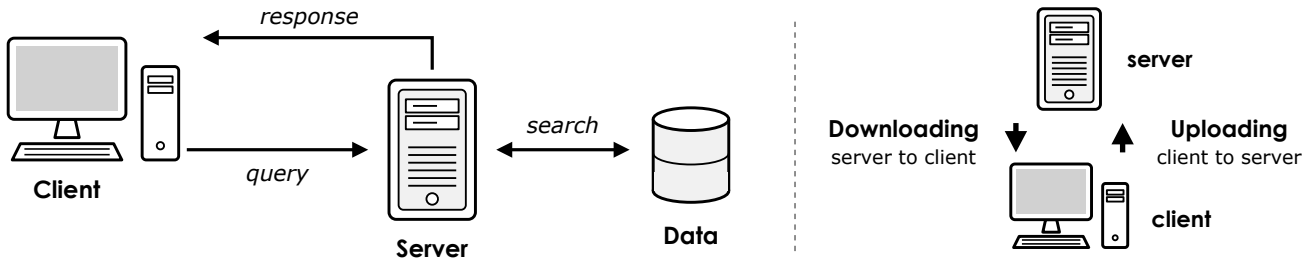


Figure 3.4 - client server architecture in the Internet; downloading and uploading

Downloading is getting information from server computers to client computers; **uploading** is giving information from client computers to server computers. The transfer is **two-way**.

Internet server	What it does
Web server	Stores web pages and provides them to the client computers.
Mail server	Stores electronic mail and provides it to the client computers.
DNS server	Translates URLs to IP addresses.

"The DNS turns names into numbers. Hard work turns dreams into results."

- THE KING OF ICT

A Put the steps in order

6 marks

How does a web page open? Number the steps 1 to 6 in the boxes.

a.	☐	The browser sends an **http get** command to the web server.
b.	☐	The user opens the web browser and enters the URL.
c.	☐	The web browser renders (shows) the HTML web page.
d.	☐	The browser now has the IP address.
e.	☐	The web server returns the HTML data stream.
f.	☐	The resolver (in the operating system) gets the IP address from the DNS server.

B Be the DNS server

4 marks

Use the table of the DNS server. Fill in the empty boxes.

The DNS server's table

Domain name	IP address
www.volusion.com	69.49.191.216
www.google.com	74.125.20.103
www.facebook.com	173.252.110.27
www.yahoo.com	209.191.122.70

The browser asks for	The DNS server replies
www.google.com	
www.yahoo.com	
www.facebook.com	
	69.49.191.216

C Which protocol?

5 marks

Write the protocol that is used: HTTP, TCP/IP, FTP, SMTP or ICMP.

What happens	Protocol
1. You open a web page: an HTML document comes to your browser.	
2. You send an e-mail to a friend.	
3. A large file is sent from one computer to another.	
4. An error happens and a message reports it.	
5. The exchange of IP addresses is controlled.	

D Downloading or uploading?

6 marks

Tick one box for each sentence.

What happens	Downloading	Uploading
Nimal saves an e-book from edupub.gov.lk on his computer.	☐	☐
Kavindi sends her project file to the school server.	☐	☐
A song from a web site is saved on a phone.	☐	☐
A teacher puts the marks sheet on the school web server.	☐	☐
You get the O/L ICT syllabus file from nie.lk.	☐	☐
A file goes from a client computer to an FTP server.	☐	☐

CHAPTER 3

Services of the Internet

? WHERE IS THE O/L SYLLABUS?

"Where is the O/L ICT syllabus on the Web?"

"Use a search engine: type a few key words!"

The Internet gives us many **services**: the WWW, e-mail, FTP, remote access, file sharing, streaming of media and search engines.



3.8 The services of the Internet

These are some of the services of the Internet (textbook page 69):

World Wide Web (WWW) | Electronic mail (e-mail) | File Transfer Protocol (FTP) | Remote access | File sharing | Streaming of media | Search engines

3.9 The World Wide Web

The **World Wide Web (WWW)** is a service of the Internet: a large collection of **electronic documents** saved in computers all around the world. **Sir Tim Berners-Lee** is the **father of the WWW**. We visit web sites with a **web browser**, an application software (Figure 3.5):

Internet Explorer | Mozilla Firefox | Google Chrome | Safari | Opera

A **web site** has several **web pages** with text, pictures, videos and **hyperlinks**; hyperlinks make browsing easy. Type the **URL** in the **address bar** and press Enter. The **home page** has the basic information about the web site; its hyperlinks open the other pages.

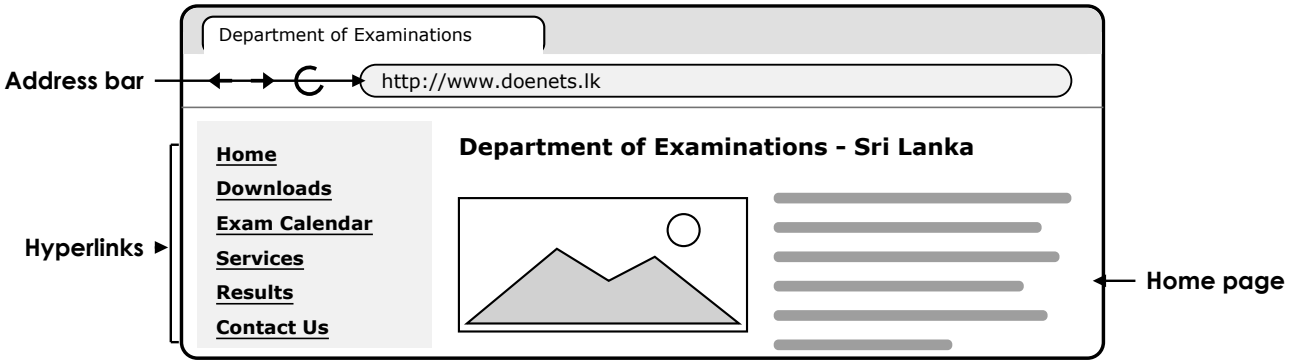


Figure 3.6 - the web site of the Department of Examinations, Sri Lanka

"A good search starts with good key words. A good answer starts with a good question."

- THE KING OF ICT

WRITING SPACE 6 - The World Wide Web write what your teacher tells you

3.10 FTP and remote access

A small file can go with an e-mail as an **attachment**, but large files cannot. **FTP** transfers **large files** from one computer in the Internet to another. It works on **TCP/IP**: it uploads the files of a client computer and downloads files from server computers.

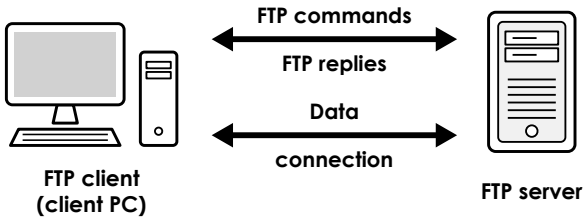


Figure 3.7 - file transfer system

Remote access controls and operates a computer in the Internet from far away (remotely), e.g. to **install software** of server computers to client computers and to **fix problems** on client computers.

BEWARE - HACKING

Hackers sometimes use remote access to steal important data and information from computers without the owner's knowledge. This is called **hacking**.

3.11 File sharing and streaming of media

File sharing is part of sharing resources: when you check exam results online, the same file is used by many clients at the same time. The Internet also shares **multimedia** - text, images, audio and video. Figure 3.9 shows the basic **streaming** process:

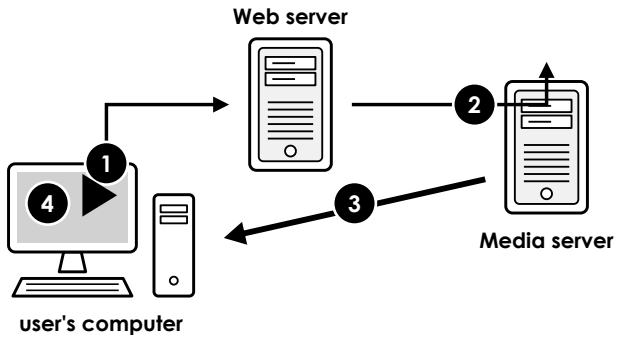


Figure 3.9 - streaming of media

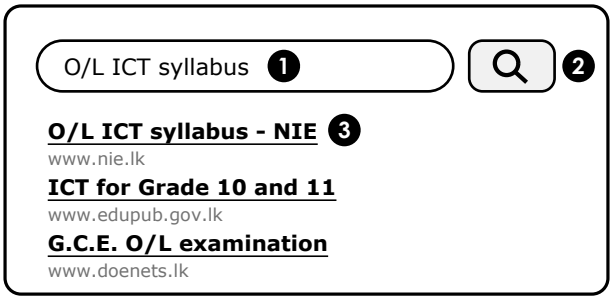
1. A user visits a web page on a **web server** and finds a file to see or hear.
2. The web server sends a message to the **streaming media server**, asking for the file.
3. The streaming server streams the file to the user's computer, **bypassing** the web server.
4. The **client software** on the user's computer decodes and plays the file.

"Never use your skills to enter another person's computer. Real skill is honest."

- THE KING OF ICT

3.12 Search engines

To open a web site you must know its URL, but the number of web sites is **unlimited**. **Search engines** help Internet users find any information easily:



1. Type the **key words** in the search box of the search engine.
2. Click the **search button**. You get a large number of web sites with hyperlinks.
3. Click one or more **hyperlinks** to get the information you need.

www.google.com / .lk www.yahoo.com www.ask.com www.msn.com

WRITING SPACE 7 - Search engines

write what your teacher tells you

DO IT - PRACTICAL 2 (TEXTBOOK PAGES 70 AND 73)

Browse a web site and search the Web. **Ask your teacher first.**

1	Open a web browser. Type **http://www.doenets.lk** in the address bar and press **Enter**.
2	Find the **address bar**, the **home page** and the **hyperlinks** (Figure 3.6).
3	Click a hyperlink to open another page. Then go back to the home page.
4	Open **http://www.google.lk**. Type the key words **O/L ICT syllabus** and click the search button.
5	Click a hyperlink in the list. Search again in **www.yahoo.com**: are the results the same?

A True or false

6 marks

Write T or F in the box.

1. ☐ A web site has only one web page.
2. ☐ Large files can always be sent as e-mail attachments.
3. ☐ FTP can upload the files of a client computer to the Internet.
4. ☐ Remote access can be used to fix problems on client computers.
5. ☐ In streaming, the media server sends the file through the web server.
6. ☐ A search engine gives a list of web sites with hyperlinks.

B Which service is it?

7 marks

Write the service of the Internet. Use the words in the box.

WORD BANK

E-mail	File sharing	FTP	Remote access
Search engines	Streaming of media	World Wide Web	

What happens	Service
1. Kasun reads web pages about the history of Sri Lanka.	
2. Dilani sends a short letter to her pen friend in Japan.	
3. A very large video file goes from one computer to another.	
4. A technician in Colombo fixes a problem on a computer in Jaffna.	
5. Thousands of students open the same exam results file.	
6. A song from a media server plays on Nimal's computer.	
7. Sanduni types key words to find web sites about elephants.	

C Web browser or search engine?

8 marks

Tick one box for each one.

Name	Web browser	Search engine
Google Chrome	☐	☐
www.google.lk	☐	☐
Mozilla Firefox	☐	☐
www.yahoo.com	☐	☐

Name	Web browser	Search engine
Safari	☐	☐
www.ask.com	☐	☐
Opera	☐	☐
www.msn.com	☐	☐

D Word search

10 marks

Find the 10 words. They go across, down and slanting - forwards and backwards.

Y	C	H	O	M	E	P	A	G	E	T
G	N	I	R	A	H	S	P	H	H	T
P	W	E	W	H	B	D	A	A	H	S
H	C	R	A	E	S	R	I	C	Y	T
G	W	N	H	D	P	O	A	K	P	R
N	W	E	A	O	C	W	K	I	E	E
U	B	K	B	N	I	Y	P	N	R	A
N	L	C	S	S	F	E	F	G	L	M
D	O	G	M	P	I	K	R	I	I	I
P	O	R	E	M	O	T	E	N	N	N
A	R	B	R	O	W	S	E	R	K	G

FIND THESE WORDS - TICK EACH ONE

- ☐ BROWSER
- ☐ HYPERLINK
- ☐ WEBSITE
- ☐ HOMEPAGE
- ☐ STREAMING
- ☐ SEARCH
- ☐ KEYWORDS
- ☐ REMOTE
- ☐ HACKING
- ☐ SHARING

CHAPTER 4

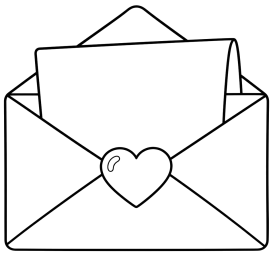
Electronic Mail

? ONE E-MAIL, THREE FRIENDS

"I want to send my results to three friends."

"Use To, Cc and Bcc, and attach the results!"

E-mail is the **cheapest** and **fastest** way to communicate. Learn the e-mail **address**, how to **send** an e-mail, and the **folders**.



3.13 What is electronic mail?

Electronic mail (e-mail) lets two or more people exchange messages and files. Many organisations give e-mail service **free of charge**, so the only cost is the use of the Internet. It is the **cheapest and fastest** communication method in the world.

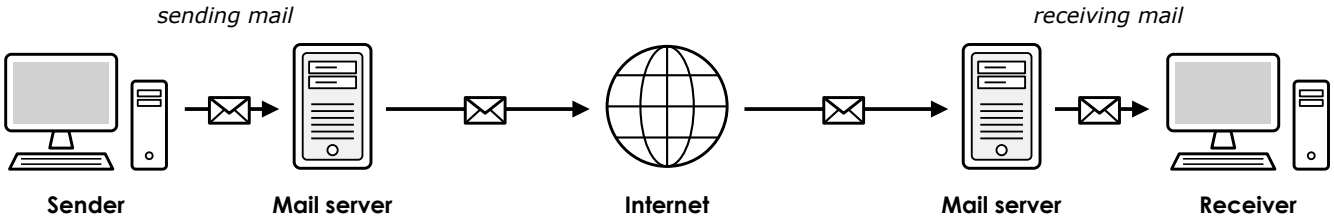
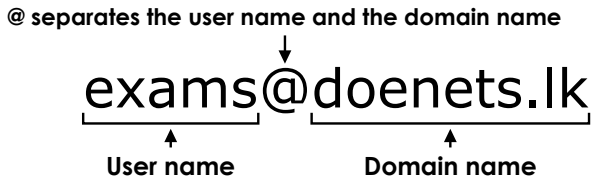


Figure 3.11 - exchange of e-mails in the Internet

3.14 The e-mail address and the account

Anyone in the world can create an **e-mail account** through the Internet; it identifies that person on the Internet. The e-mail address of the Department of Examinations:



exams is the **user name**, **doenets.lk** is the **domain name**, and the **@** symbol separates them. The sender and the receiver must both have e-mail accounts. An account has an **e-mail address** and a **password**: to open it, type the user name and the password exactly and click **Sign In**.

WRITING SPACE 8 - E-mail addresses write what your teacher tells you

3.15 Sending an e-mail

Shani sends her O/L results to three friends (Figure 3.12):

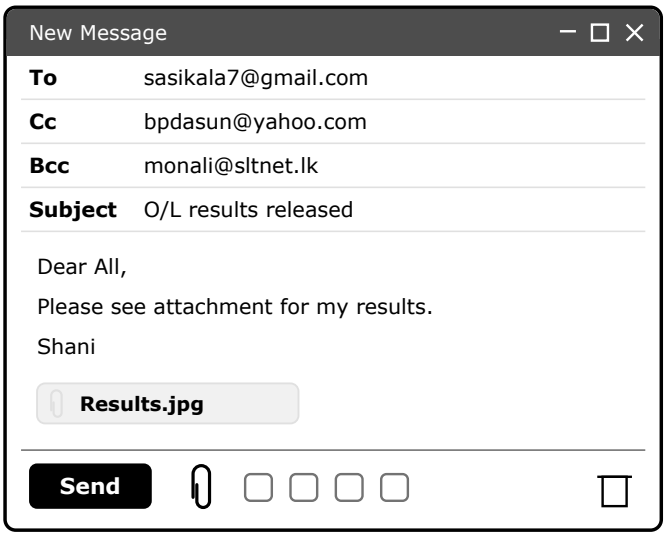


Figure 3.12 - sending an e-mail

- **To** - The receiver's e-mail address (sasikala7@gmail.com).
- **Cc** - **Carbon copy**: others who should get a copy (bpdasun@yahoo.com).
- **Bcc** - **Blind carbon copy**: receivers that the To and Cc receivers do not know about (monali@sltnet.lk).
- **Subject** - The title of the mail (O/L results released).
- **Attachment** - The button with the **paper clip** icon attaches files (Results.jpg).
- **Send** - Click **Send** to send the e-mail.

WHO CAN SEE WHOM?

The **Bcc** receiver (monali@sltnet.lk) can see all the addresses the mail was sent to. The **To** and **Cc** receivers (sasikala7@gmail.com and bpdasun@yahoo.com) **cannot** see the Bcc address.

WRITING SPACE 9 - To, Cc and Bcc

write what your teacher tells you

3.16 E-mail folders and spam

E-mails are kept in **folders** for easy use. **Spam** is unwanted e-mail stored in a separate folder, so that it does not disturb the Inbox (Figure 3.13).

	Folder	What it stores
	Inbox	Stores the mails you received.
	Drafts	Stores mails that were composed but could not be completed.
	Sent	Stores the mails you sent.
	Trash / Deleted	Stores deleted mails for a certain period.
	Spam / Junk	Stores the mails that are unwanted.

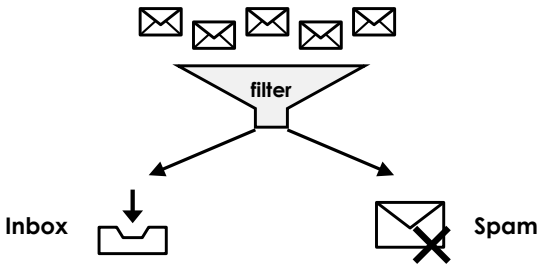


Figure 3.13 - spam mail

DO IT - PRACTICAL 3 (TEXTBOOK PAGES 74 - 76)

Use the e-mail account your teacher gives you. **Never share your password.**

1	Open the e-mail web site. Type your **user name** and **password**. Click **Sign In**.
2	Click **Compose**. Type a friend in **To**, another in **Cc** and your teacher in **Bcc**.
3	Type a **Subject** (e.g. My ICT homework) and a short, polite message.
4	Click the **paper clip**, attach a small picture and click **Send**.
5	Find your mail in **Sent**. In the **Inbox**, open your friend's mail: is its Bcc shown?
6	Close a new mail without sending it: find it in **Drafts**. Look at **Trash** and **Spam**.
7	Do not open unknown mails in Spam. Click **Sign out** at the end.

A Parts of e-mail addresses

8 marks

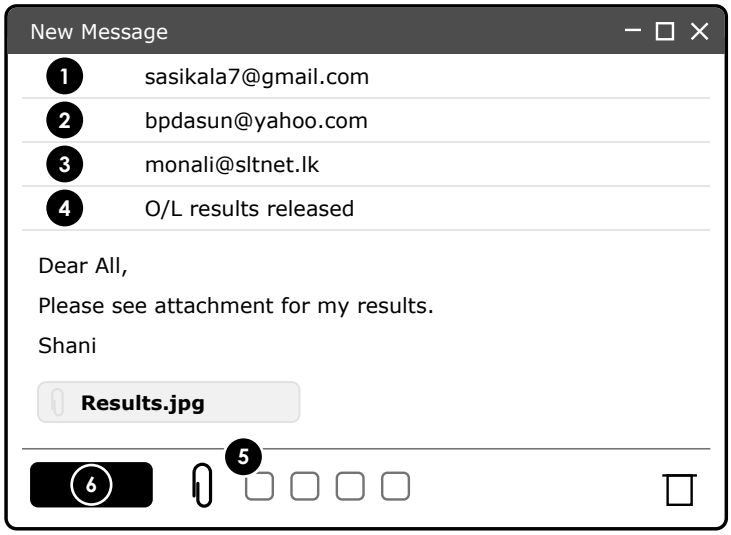
Write the user name and the domain name.

E-mail address	User name	Domain name
sasikala7@gmail.com		
bpdasun@yahoo.com		
monali@sltnet.lk		
somasiri@gmail.com		

B Label the e-mail window

6 marks

Write the names of the numbered parts 1 - 6.



USE THESE WORDS

Attachment	Bcc
Cc	Send
Subject	To

1. .....
2. .....
3. .....
4. .....
5. .....
6. .....

"Write every e-mail as if your teacher will read it."

- THE KING OF ICT

C Who can see the addresses?

6 marks

Shani's e-mail goes To Sasikala, Cc Dasun and Bcc Monali. Tick Yes or No.

Question	Yes	No
Can Sasikala (To) see Dasun's (Cc) address?	☐	☐
Can Dasun (Cc) see Monali's (Bcc) address?	☐	☐
Can Monali (Bcc) see Sasikala's (To) address?	☐	☐
Can Sasikala (To) see Monali's (Bcc) address?	☐	☐
Can Monali (Bcc) see Dasun's (Cc) address?	☐	☐
Do all three friends get the e-mail?	☐	☐

D Which folder?

5 marks

Where is each mail? Write Inbox, Drafts, Sent, Trash or Spam.

The mail	Folder
1. A mail your friend sent you this morning.	
2. A mail you started to write but could not finish.	
3. A mail you sent to your teacher yesterday.	
4. A mail you deleted last week.	
5. An unwanted mail from an unknown sender.	

CHAPTER 5

Communication and the Cloud

? A LESSON FROM FAR AWAY

"Our ICT teacher is in Kandy today!"

"She can still teach us - by video conference."

Learn about **instant messages**, **video conferencing**, **social networks** and **cloud computing**.



3.17 Instant messages

With the **chat** facility, two persons exchange **short messages very fast** over the Internet: **instant messages (IM)**. A member message can reach a member's computer, phone (as voice) or fax (Figure 3.14). Tools (Figure 3.15): WhatsApp, Viber, Skype, LINE, WeChat and Messenger.

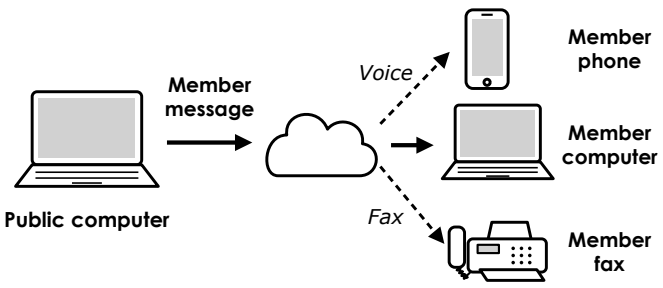
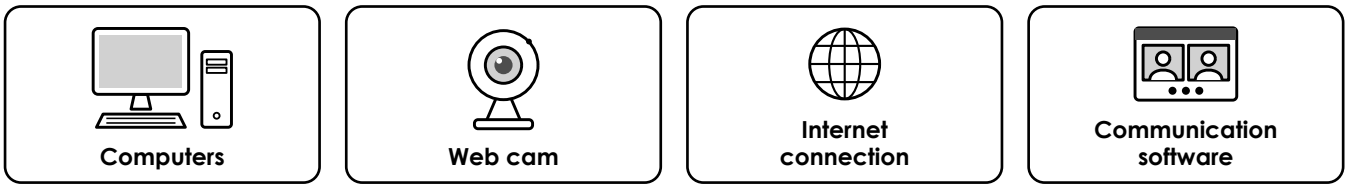


Figure 3.14 - sending instant messages

3.18 Video conference

Video conferencing is a conversation between two or more persons in **different (remote) locations**, using **audio and video** transmission over computer networks. It needs:



The communication software must have **audiovisual** facilities (sound and pictures).

WRITING SPACE 10 - Video conference write what your teacher tells you

3.19 Social networks

Social networks maintain **social relationships** over the Internet: people exchange personal details, photographs, videos and daily activities. Relationships are **widened**, but social networks can also create **social issues**. Examples:

www.facebook.com | www.twitter.com | www.youtube.com
www.flickr.com | www.pinterest.com | www.secondlife.com

3.20 Cloud computing

Cloud computing uses computer networks, including personal computers and mobile devices, to **retrieve and store information from anywhere at any time**. Its three main services:

Service	What it provides	Example
IaaS Infrastructure as a Service	A virtual environment of servers from large data centres: space to store data and software applications, and other resources.	Server space and a firewall through cloud computing, though you do not own them.
PaaS Platform as a Service	The server environment for software development: the operating system, a programming language environment, a database and a web server.	Develop software in the cloud with no operating system or compilers installed on your computer.
SaaS Software as a Service	Software installed in the cloud, used without installing it. It cuts the cost and complexity of buying and managing hardware and software.	Word processing and spreadsheet software used through a web browser on an Android device.

"Connect with people who make you better, online and offline."

- THE KING OF ICT

ADVANTAGES

- Lower software cost
- Improve performance
- Fewer maintenance issues
- Instant software updates

DISADVANTAGES

- Requires a constant Internet connection
- Does not work well with low speed Internet connections

WRITING SPACE 11 - Cloud computing

write what your teacher tells you

DO IT - PRACTICAL 4 (TEXTBOOK ACTIVITY, PAGE 80)

Find each item of the textbook Activity on a computer. Then match the columns in the exercise **Match A and B**.

1	Open the web browser **Mozilla Firefox** (or Google Chrome).
2	Type the URL **http://www.nie.lk** in the address bar. Point to its domain name.
3	Open the search engine **www.google.lk**. Search for **Department of Examinations**.
4	Open www.doenets.lk from the results. Find an e-mail address on its contact page.
5	Look at the IP address of your computer again (Practical 1): four values and dots.

A Match A and B

8 marks

The textbook Activity (page 80). Write the correct letter in the box.

A column		B column
1. E-mail address		**A.** 172.92.83.106
2. Domain name		**B.** Converts domain names into IP addresses
3. Uniform Resource Locator		**C.** somasiri@gmail.com
4. IP address		**D.** Google
5. Web browser		**E.** doenets.lk
6. A service of the Internet		**F.** World Wide Web
7. Domain Name Server		**G.** http://www.nie.lk/syllabus/ol/ict.pdf
8. Search engine		**H.** Mozilla Firefox

"Share your good moments online, but keep your private life private."

- THE KING OF ICT

B Which way to communicate?

6 marks

Tick one box for each sentence.

What people do	E-mail	Instant message	Video conference	Social network
Two friends chat with short messages very fast.	☐	☐	☐	☐
Doctors in two countries talk with web cams.	☐	☐	☐	☐
Kamal shares his trip photos with his relatives.	☐	☐	☐	☐
A school sends a letter and a file to a friend school.	☐	☐	☐	☐
Principals in five districts meet with audio and video.	☐	☐	☐	☐
Nimali follows her old classmates' daily activities.	☐	☐	☐	☐

C IaaS, PaaS or SaaS?

6 marks

Tick the cloud service that is used.

In the cloud ...	IaaS	PaaS	SaaS
A company uses server space and a firewall from the cloud.	☐	☐	☐
Software is developed in the cloud without compilers.	☐	☐	☐
A word processor is used in a browser, not installed.	☐	☐	☐
Data centres give space to store data and software.	☐	☐	☐
An OS, a database and a web server for developers.	☐	☐	☐
A spreadsheet is used in a browser on an Android phone.	☐	☐	☐

D Answer these

4 marks

Write full sentences.

1. Write the four requirements of a video conference. (2 marks)

2. Write one good result and one problem of social networks. (2 marks)

"Your files can live in the cloud; keep your feet on the ground."

- THE KING OF ICT

REVISION

Unit 3 in One Page

Read this the night before the test

Topic	The key facts
The Internet	A collection of computer networks around the world - the information super highway. No single owner; the Internet Society (non-profit) looks after its ethics, principles and protocols.
URL	Uniquely identifies a resource in a web site: http:// (protocol) www. (WWW) edupub.gov.lk (domain name) /e-books/english/ (path) ict.pdf (resource). It must have the parts from HTTP to the domain name.
IP address	Identifies each computer in the Internet: four values from 0 to 255 separated by dots (dotted decimal notation), e.g. 172.64.85.42. Given by an ISP (a company that charges fees).
Domain name	The identity of a web site: com, org, gov, edu, net; lk, in, au, jp, uk, us. The right-most domain is the Top Level Domain (lk in edupub.gov.lk).
DNS	The Domain Name System (Server) translates a domain name into an IP address: www.yahoo.com → 209.191.122.70. The connection is made with the IP address.
Protocols	A system of law: HTTP (HTML documents), TCP/IP (IP addresses), FTP (files), SMTP (e-mail), ICMP (messages about errors).
Client server	The Internet is a WAN. Downloading: server to client; uploading: client to server. Web, mail and DNS servers.
Services	WWW (father: Sir Tim Berners-Lee; browsers; home page; hyperlinks), e-mail, FTP, remote access (hacking!), file sharing, streaming of media, search engines (key words).
E-mail	Cheapest and fastest. user name @ domain name. To, Cc, Bcc, Subject, Attachment (paper clip), Send. Inbox, Drafts, Sent, Trash, Spam.
Communication	Instant messages (chat); video conference (computers, web cam, Internet connection, communication software); social networks (relationships widen, social issues).
Cloud computing	Store and retrieve information from anywhere at any time. IaaS, PaaS, SaaS. Lower software cost, instant updates; needs a constant Internet connection.

TICK WHEN YOU CAN DO IT

- [ ] I can explain what the Internet is and who looks after it.
- [ ] I can name the five parts of a URL and find the Top Level Domain.
- [ ] I can say whether an IP address is correct and who gives it.
- [ ] I can explain how the DNS, the protocols and the servers work.
- [ ] I can name the seven services of the Internet and use a search engine.
- [ ] I can send an e-mail with Cc, Bcc, a subject and an attachment.
- [ ] I can explain IM, video conferencing, social networks and the cloud.

WORD POWER

Unit 3 Word List

Thirty-two words you must know

Word	What it means
Internet	A collection of computer networks around the world.
Internet Society	Non-profit; in charge of the ethics and protocols.
URL	Uniform Resource Locator: identifies a web resource.
IP address	Identifies a computer: four values from 0 to 255.
Dotted decimal notation	An IP address written as four values with dots.
ISP	Internet Service Provider: a company, charges fees.
Domain name	The identity of a web site, e.g. edupub.gov.lk.
Top Level Domain	The right-most domain, e.g. lk, com, org.
DNS	Domain Name System: turns names into IP addresses.
Protocol	A system of law in networks, e.g. HTTP, FTP, SMTP.
Client	A computer that gets resources from a server.
Server	A computer that gives resources to the clients.
Downloading	Getting information from a server to a client.
Uploading	Giving information from a client to a server.
World Wide Web	A large collection of electronic documents.
Web browser	Application software that opens web sites.
Home page	The page with the basic information of a site.
Hyperlink	A link that opens another page; easy browsing.
Remote access	Controlling a computer in the Internet remotely.
Hacking	Stealing data without the owner's knowledge.
Streaming	A media server streams a file for the user to play.
Search engine	Finds information from key words, e.g. google.lk.
Bcc	Blind carbon copy: hidden from the To and Cc people.
Spam	Unwanted e-mail stored in a separate folder.
Video conference	Talking with audio and video across locations.
Cloud computing	Storing and getting information anywhere, any time.
IaaS / PaaS / SaaS	Infrastructure / Platform / Software as a Service.
HTTP	Hypertext Transfer Protocol: HTML document exchange.
FTP	File Transfer Protocol: transfers large files.
SMTP	Simple Mail Transfer Protocol: e-mail exchange.
Attachment	A file sent with an e-mail (the paper clip button).
Instant message	A short, very fast message sent with the chat facility.

"Know the words of the Internet, and every web page will speak to you."

- THE KING OF ICT

END OF UNIT

Unit 3 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. Which one is a correct IP address?
   (a) 192.168.2.10 (b) 192.168.256.10 (c) 192.168.2 (d) 192.168.2.10.1
2. In http://www.nie.lk/pages/syllabus.asp the resource is
   (a) http (b) nie.lk (c) /pages/ (d) syllabus.asp
3. Which protocol is used for electronic mail exchange?
   (a) HTTP (b) FTP (c) SMTP (d) ICMP
4. A domain name is converted into an IP address by the
   (a) web server (b) mail server (c) DNS server (d) ISP
5. Getting information from a server to a client is
   (a) uploading (b) downloading (c) streaming (d) hacking
6. Which one is a web browser?
   (a) www.google.lk (b) www.yahoo.com
   (c) Mozilla Firefox (d) www.msn.com
7. The receivers who are hidden from the others are typed in
   (a) To (b) Cc (c) Bcc (d) Subject
8. A - SaaS lets you use software in the cloud without installing it. B - A video conference needs a web cam. C - The mails you could not complete are stored in the Sent folder. Which are correct?
   (a) A and B only (b) A and C only (c) B and C only (d) A, B and C

2 Fill in the blanks

8 marks

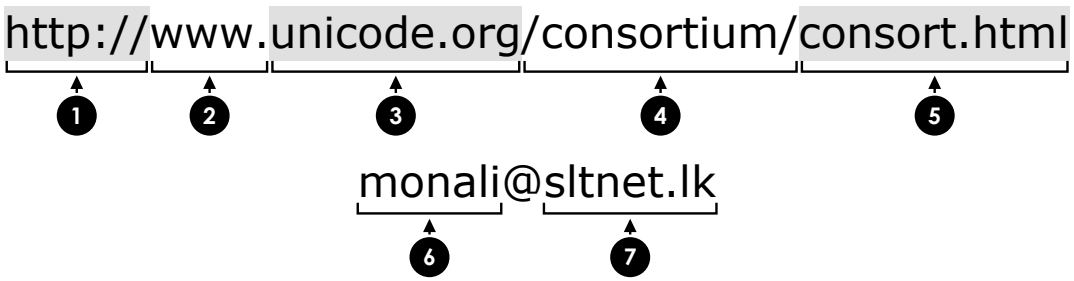
One word in each blank.

1. The non-profit Internet ..... looks after the protocols of the Internet.
2. An IP address has four values from 0 to ..... .
3. The right-most domain of a domain name is the Top ..... Domain.
4. ..... is the protocol for HTML document exchange.
5. Large files go from one computer to another with ..... .
6. The page with the basic information of a web site is the ..... page.
7. In exams@doenets.lk, doenets.lk is the ..... name.
8. Unwanted e-mails kept in a separate folder are called ..... .

3 Look at the URL and the e-mail address

8 marks

Name the parts 1 - 7. In 8, write the Top Level Domain of the URL.



1. ..... 2. .....

3. ..... 4. .....

5. ..... 6. .....

7. ..... 8. .....

4 Answer these

16 marks

Write in the space given.

1. What is an IP address? Who gives IP addresses? (2 marks)

2. Explain how the DNS helps a web browser to open www.yahoo.com. (3 marks)

3. Name three Internet servers and write what each one does. (3 marks)

4. Write the four requirements of a video conference. (2 marks)

5. Shani e-mails To Sasikala, Cc Dasun, Bcc Monali. (a) Who cannot see Monali's address? (b) How is a file sent with it? (c) Which folder keeps the sent mail? (3 marks)

6. What is Software as a Service (SaaS)? Give one example. (3 marks)